

TECH LEAD HANDOVER: AI-POWERED HRMS CORE SPECIFICATION

PHASE 1: SYSTEM ARCHITECTURE FRAMEWORK (FOUNDATION)

Developers sab se pehle is base structural framework par kaam shuru karenge:

- **Multi-Company SaaS Engine (Module 27, 28):** Multi-tenant database design jahan har company ka data complete isolate ho. Stripe/Paddle integration for international monthly subscription handling.
- **Workflow Automation Builder (Module 30):** Centralized Rule Engine (e.g., If Leave_Approved -> Then Trigger_Notification & Update_Attendance).
- **Security & Anti-Fraud Layer (Module 3, 22):** Hooks jo mobile application endpoints par Request Headers check karein (VPN, Rooted Device, Fake GPS Mock Providers detection).

PHASE 2: MODULE-WISE TECHNICAL BREAKDOWN

1. Unified Dashboard & Analytics Hub (Modules 1, 10, 29)

- **Data Aggregation Engine:** Redis caching layer taake Live Workforce tracking aur analytics dashboard par query load kam ho.
- **AI Integration:** LLM integration with database views to provide **AI Predictive Analytics** (Resign risk prediction matrix using employee activity data and historical turnover patterns).
- **Key Views:** Charts.js/D3.js dynamic graphs for financial, recruitment, and productivity KPIs.

2. Core Employee & Identity Management (Modules 2, 21, 31, 32)

- **Employee Ledger:** Complete chronological timeline mapping employee transfers, promotions, and contracts.
- **Document OCR Layer:** Integration with Vision models/OCR APIs for **CNIC parsing, Passport data extraction, and CV parsing** to auto-fill onboarding fields.
- **E-Signature Sandbox:** Cryptographically secured digital signatures for contracts and offer letters with audit logs.

3. High-Security Advanced Attendance System (Module 3 - Onsite/Remote/Outdoor)

- **Onsite API:** Webhook for Biometric hardware integrations. Mobile front-end QR generator with a rolling 5-second expiration token (Screenshot protection).
- **Outdoor & Geofencing:** Google Maps API/Mapbox integration for perimeter bounding (Geofencing) and route historical tracking.
- **AI Selfie Verification:** Face API/AWS Rekognition integration to match attendance selfies with the baseline profile photo to avoid proxy attendance.

4. Productivity, Time Tracking & Monitoring (Modules 8, 14, 26)

- **Agent Desktop Client:** A lightweight background application (C# / Electron) for desktop activity monitoring:
 - Mouse/Keyboard velocity logs (Idle detection threshold: 5 mins).
 - Random screenshot capturing pipeline (compressed and stored securely in AWS S3 buckets).
- **Shift Engine:** Automated roster scheduler with logical fallback constraints (e.g., Minimum 11 hours rest between rotational shifts).

5. Talent Acquisition & AI Recruitment System (Modules 6, 31)

- **AI Pipeline:**
 - **CV Matching:** Vector embeddings match candidate profiles against the job description vectors using a Vector Database (e.g., Pinecone / PGVector).
 - **AI Interview Agent:** Generates targeted technical questions based on the candidate's specific resume gaps.

6. Compensation, Finance & Workflow Operations (Modules 5, 23, 24)

- **Payroll Engine:** Highly configurable mathematical engine handling variable deductions (Late arrivals, unpaid leaves), bonuses, overtime multipliers, and localized income tax brackets.
- **Financial Claims System:** Optical scanning of travel/fuel receipts linked directly to expense approval workflows.

7. Core AI Assistant & Internal Collaboration (Modules 13, 15, 16)

- **Contextual HR Chatbot:**
 - *Employee Scope:* Retrieval-Augmented Generation (RAG) system connected to **Module 32 (Company Knowledge Base)** so employees can ask chatbot about policies, remaining leaves, and salary status.
 - *Manager Scope:* Multi-turn data querying (e.g., "*Who was late more than 3 times this month?*" converts natural language to SQL query).

- **Communication Layer:** WebSockets (Socket.io) for real-time employee chat, file sharing encryption, and WebRTC for peer-to-peer audio/video calls.

8. Secondary Operational Modules (Modules 4, 7, 9, 11, 12, 17, 18, 19, 20, 25)

- **LMS:** Video streaming content storage, quiz engine, and automated certificate generation logic.
 - **Asset Lifecycle Tracker:** Fixed-asset depreciation logs and digital sign-off records upon equipment assignment.
 - **Visitor Entry Gateway:** Self-service QR generation kiosk app for scanning visitor CNICs/IDs and notifying the host employee instantly.
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PRODUCT REQUIREMENT DOCUMENT (PRD) / TECH HANDOVER

Project Name: AI-Powered Revenue Intelligence, CRM & Field Force Ecosystem

Target Audience: Global B2B SaaS Market (SMBs, Retail, Institutes, Enterprises)

Architecture: Multi-Tenant Micro-services Framework with Dynamic UI Localization

PHASE 1: CORE ARCHITECTURE & DATA PIPELINES (FOUNDATION)

Developers core system architecture design karte waqt sab se pehle in base pipelines ko configure karenge:

- **Multi-Pipeline Event Engine:** Multi-tenant database schema jahan har tenant multiple custom sales pipelines (e.g., Inbound Sales, Outbound Sales, Institutional Admissions) drag-and-drop Kanban state machine ke zariye handle kar sake.
- **Omnichannel Communication Router:** Centralized database table jo Email sync (IMAP/SMTP), WhatsApp Business API webhooks, SMS gateways, LinkedIn messaging integrations, aur Call Logs (Twilio/Exotel) ko combine kar ke har lead profile par ek single chronological timeline feed banaye.
- **Telemetry & Anti-Fraud Engine:** Real-time GPS coordinates stream handler (Redis Geo commands ke sath) jo mobile applications se telemetry data le kar spoofed/mock locations, VPNs, aur rooted devices ko track aur drop kare.
- **Dynamic UI Translation Engine:** Core localization layer jo database se tenant ki industry metadata read karegi aur pure CRM ke labels (Lead, Product, Delivery) ko industry-specific terminology mein switch karega.

PHASE 2: MODULE-WISE TECHNICAL BREAKDOWN

1. Unified Revenue Intelligence Dashboard (Module 1)

- **Data Aggregation Engine:** Redis caching layer taake live workforce tracking, total sales metrics, revenue overview, aur conversion analytics ka data main database par heavy queries run kiye bina real-time update ho.
- **AI Analytics Layer:** Past velocity aur sales trends ko evaluate kar ke monthly/yearly growth charts aur charts.js/D3.js ke dynamic graphs compile karna.

2. Marketing Hub & Ad Campaign Command Center (Module 14)

- **Ad Ingestion Webhooks:** Direct integrations with Meta Graph API (Facebook & Instagram Lead Ads), Google Ads API, and LinkedIn Lead Gen Forms.
- **Automated Lead Routing Engine:** Jaise hi third-party platform se lead ingest ho, system tenant-defined routing rules execute kare:
 - *Round-Robin Routing:* Specific sales agents ke dashboards par leads ka barabar batwara.
 - *Direct WhatsApp Dispatch:* Lead alert automated message text mein transform ho kar relevant department ke internal WhatsApp group mein directly webhook ke zariye chala jaye.
- **Social Media Campaign Scheduler:** ERP ke andar se hi graphics aur content directly Meta aur LinkedIn platforms par posts schedule aur publish karne ke API endpoints.

3. Smart Lead Management & Duplicate Detection (Module 2)

- **AI Lead Scoring Router:** Lead creation par system source, geography, industry size, aur metadata analyze kar ke automatic priority scoring assignment (1-100) karega.
- **Fuzzy-Matching Duplicate Detector:** Algorithm jo incoming leads ke phone numbers, email domains, aur company names ko scan kare taake different internal sales reps ke darmiyan ownership conflicts na hon.

4. Omnichannel Agent Portal & Post-Sale Lifecycle (Modules 3, 5, 7, 20)

- **Interaction Outcome Logger:** Sales rep jab client ko contact kare (Call, WhatsApp, Email, ya AI Bot), to is portal par details enter karega:
 - *Client Status Toggles:* Interested, Not Interested, Call Back, Busy.
 - *Follow-up Scheduler:* Logs next actions, dates, aur precise timestamps.
 - *Rich Text Call Notes:* Call par hui poori baat ka detail summary input box.
- **Post-Sale Fulfillment Tracking:** Agar deal status "Won/Sold" ho jaye, to screen par conditional operations metadata forms render hon:
 - *E-Commerce/Retail:* SKU/Variant Selection, Final Sold Price, Courier API payload initialization (Self-Pickup vs Delivery Mode).
 - *Education / Training Institutes:* Batch Start/End Dates, Class Timings, Assigned Instructor/Trainer Profiles.
- **Online Business Ingestion webhooks:** Pre-built modules for WooCommerce, Shopify, and custom REST endpoints taake online businesses ka external check-out data directly is portal par bina manual entry ke sync ho sake.

5. Industry-Specific Dynamic UI Localization (USP Feature)

- **Terminology Mapping Layer:** System interface par elements static strings nahi honge balki data-attributes se bind honge. Signup ke waqt company jo industry select karegi, UI usi terminology ke labels load karega:

Industry Meta Type	Element 1 (Lead Label)	Element 2 (Product Label)	Element 3 (Fulfillment Label)
Education / Institute	Student Name	Course / Program	Batch Start Date & Class Timing
Retail / E-Commerce	Customer Name	Item / SKU / Variant	Courier Dispatch / Courier Track
B2B / Software House	Corporate Client	Tech Service / Module	Project Kick-off / Delivery Date
Real Estate	Investor / Buyer	Property / Plot Number	Possession / Registry Date

6. Cross-Departmental Accounts Request Engine (Modules 8, 10)

- **Inter-Module Intercept:** Sales agent ko accounts module ya core ledgers ka direct read/write access nahi hoga.
- **Workflow Request Bridge:**
 1. Sales rep portal par rehte hue "Request Quotation / Invoice" trigger karega, pricing brackets aur proposed discount apply kar ke form submit karega.
 2. Accounts department ke dashboard par real-time API push notification ke sath workflow ticket alert generate hoga.
 3. Accounts manager parameters verify kar ke jaise hi approve karega, ledger update hoga aur system automatically PDF quotation/invoice generate kar ke Sales portal aur **Customer Portal (Module 29)** dono par update kar dega.

7. Actionable Notification & Proactive Reminder Engine (Module 21)

- **Cron-Job Scheduler:** Background workers continuous runs pe rahenge notifications pipeline execute karne ke liye:
 - *Sales Reminders:* Scheduled follow-ups se 15 mins pehle mobile push aur browser notifications message template payload ke sath.
 - *Billing Alerts:* Client ki subscription billing start hote hi ya invoices process hote hi custom status messages.

8. High-Performance Field Force Engine (Module 6)

- **Route Optimization TSP (Traveling Salesperson Problem) Algorithm:** Sales agent ke assigned field addresses ko analyze kar ke shortest geographical route schedule design karna Google Maps Matrix API ke zariye.
- **Geofenced Check-In:** Coordinates match execution jo verification confirm kare ke sales rep client location ke 50-meter radius mein मौजूद hai tabhi selfie verification panel trigger ho.

9. Call Center Center Integration & AI Summarization (Module 15)

- **Click-to-Call System:** Browser SDK integration VoIP outbound dialing handle karne ke liye.
- **Whisper API Transcription Pipeline:** Outbound call record hote hi, processing worker background activity execution run karega—audio file text mein convert hogi aur AI pipeline us ka concise actionable summary nikal kar interaction logs table mein insert kar dega.

10. Workflow Automation Builder & Rule Engine (Module 25)

- **Event-Condition-Action (ECA) Gateway:** Dynamic system parameters evaluate karne ke liye backend node engine:
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PRODUCT REQUIREMENT DOCUMENT (PRD) / TECH HANDOVER

Project Name: Universal Smart POS & Predictive Inventory Ecosystem (Module 3)

Performance Standard: Sub-100ms Billing Execution, Zero-Latency Keystroke Processing, Proactive AI Context Ingestion

Architecture: Hybrid Offline-First Local Storage States with Native Hardware Proxy Bridge (Electron/C#)

PHASE 1: CORE ARCHITECTURE & DATA ENGINES (FOUNDATION)

Developers core POS engine design karte waqt sab se pehle in base frameworks ko configure karenge:

- **Offline-First IndexedDB Sync Engine:** Browser memory mein Local Storage ya IndexedDB ka usage taake internet disconnect hone par bhi billing, barcodes processing, aur customer catalogs flawlessly chalti rahein. Internet restore hote hi background worker automatic delta sync pipeline run kar ke local records ko cloud database par push karega.
- **Hardware Interface Middleware Bridge:** Client device par ek native proxy wrapper service run karegi (e.g., Electron Node-HID wrapper). Yeh external peripheral hardware signals listen kar ke directly web interface container mein standard payloads inject karegi:
 - *Raw USB/Bluetooth HID Data:* Instant capture mechanism for Zebra, Honeywell, and Datalogic barcode scanners.
 - *ESC/POS Command Streaming:* Raw byte streams configuration thermal receipt printers aur RJ11 cash drawers trigger control pulse hardware lines ke liye.
 - *Serial COM Interfacing (RS232/USB):* Virtual COM ports reader weighing scales ke liye jo continuous output load metrics ko live cart counters mein stream kare.
- **Dynamic UI Translation & Industry Switching Node:** Layout metadata processor jo merchant industry settings (Retail, Restaurant, Pharmacy, Wholesale) ke mutabiq core UI viewports aur application operations rules ko on-the-fly dynamically swap/switch kare.

PHASE 2: MODULE-WISE TECHNICAL BREAKDOWN

1. High-Speed Multi-Input Billing UI Layer (Module 1, 15)

- **Keystroke Global Interceptor:** Core operational dashboard interface background listeners bind karega jo browser default keys logic (F1-F8, Ctrl+Q, etc.) ko intercept/override karein taake keyboard-only checkout speeds maintain rahein.

Functional Key	System Action Handler	UI Event / Focus Allocation
F1	initiateNewInvoice()	Clears active state and mounts a clean invoice model object.
F2	focusProductSearch()	Explicitly targets search bar input element node.
F3	openCustomerModal()	Launches slide-over canvas interface for managing customer records.
F4	holdActiveInvoice()	Dispatches active cart schema to temporary browser cache array.
F5	applyCartDiscount()	Spawns contextual override numerical keypad form.
F6	triggerPaymentWorkflow()	Shifts focus matrices straight onto alternative checkout gateways list.
F7	executeFinalPrinting()	Submits finalized raw binary payload streams to hardware bridge.

F8	<code>pulseCashDrawer()</code>	Executes raw control breakout command bytes.
Ctrl + Q	<code>modifyActiveItemQty()</code>	Instantly focuses and unlocks quantity array indexes on active row.
Del	<code>removeSelectedRow()</code>	Expunges highlighted cart index element reference from temporary collection.
Ctrl + S	<code>saveDraftInvoice()</code>	Registers current data attributes into local backup snapshot.

2. Universal Hardware & Custom Printing Matrix (Modules 2, 3, 4, 5)

- **ESC/POS Native Printer Assembler:** Dynamic template assembler engine jo text block variables, logos metadata, aur country-specific compliant QR structures (Zakat/Tax validation formats) ko binary command structures array buffer mein translate kare.
- **Label Layout Engine (ZPL/TSPL Parser):** Canvas module jo web view coordinates ko precise device dimensions format (Zebra ZPL code matrices ya TSC TSPL instructions) mein change kare taake SKU stickers, shelf stickers, aur barcodes printing standard design accuracy ke sath output hon.

3. Industry-Specific UX Engine Contexts (Module 10)

- **Supermarket/Retail:** Non-blocking scanner thread listening capability with fast checkout layouts and hotkeys loops grid configuration.
- **Restaurant/F&B:** Dynamic digital floor grids mapping table systems, split billing matrix calculators, and kitchen order ticket (KOT) multi-channel printing pipelines.
- **Pharmacy:** Expiry date tracking data blocks, batch validations array monitors, and conditional FIFO automation checks.
- **Wholesale:** Layered multi-tier tiered wholesale pricing data mapping fields customer verification status metadata ke mutabiq.

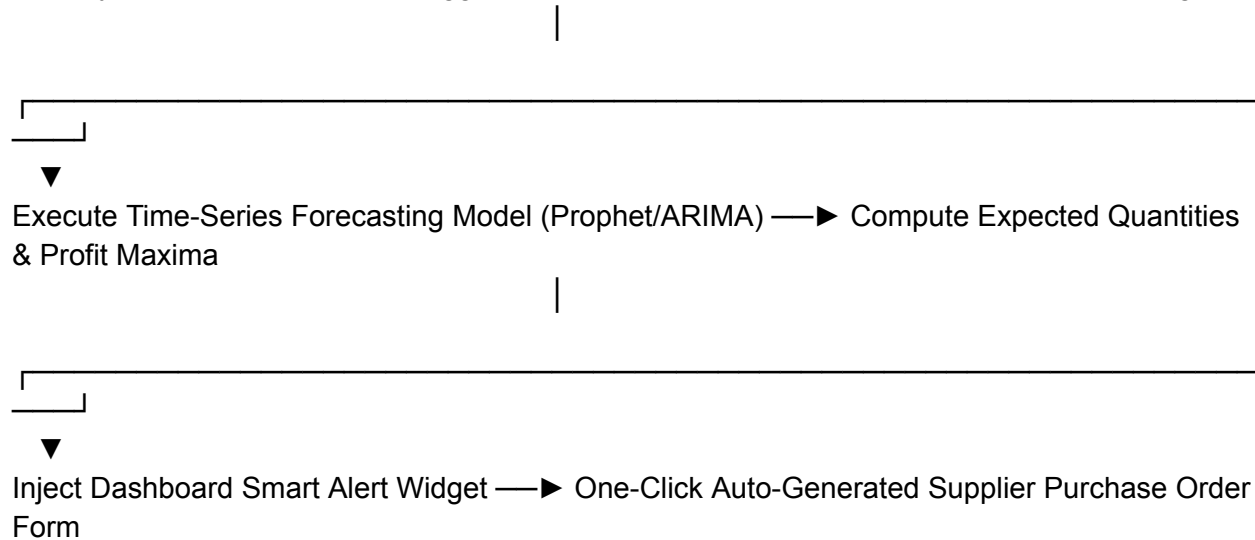
4. AI Automated Product Velocity & Smart Restocking Consultant (Module 11)

- **Dynamic Product Classifier Pipeline:** Background automated micro-service execution metrics tracker jo sales throughput volume data run aggregates scan kare aur hot-seller index mapping verify kare.
- **Proactive Stocking Advisor Engine:** AI models analyze karengi item inventory burn rates ko current status numbers ke parallel.
 - *System Action:* Agar inventory volume drops past velocity metrics curve pattern se cross kare, to dashboard alert parameters block create hoga: "Your 'Beverages' category sales increased by 35% this month. Product X is moving fast. We advise stocking at least 400 units to avoid out-of-stock leakages."

5. Occasion-Aware & Seasonal Demand Forecasting Engine (Module 11)

- **The 60-Day Look-Ahead System Scheduler (Cron Architecture):** Background service worker engine jo constant timelines evaluation run kare global/regional fixed aur shifting calendar events mapping arrays par (e.g., Eid, Christmas, Back-to-School season).

[60 Days Prior to Event] —▶ Trigger Look-Ahead Worker —▶ Pull Last Year's Sales Logs



- **Predictive Metrics Execution Target:**
 - *Historical Ingestion:* Automatic retrieval matrices filtering last year's exact calendar data parameters loop.
 - *Quantity & Profit Forecasting Engine:* Multi-variable regression parsing rules calculation metrics run kar ke merchant dashboard module frame par exact predictions structure pass karega:
 - Recommended Inventory Stock Batch Size: 850 Units
 - Predicted Gross Sales Horizon for Upcoming Event: \$15,400 (+22% YoY Shift)
 - Estimated Net Revenue Profit Window Margins: \$4,100
 - *One-Click Procurement Integration:* Action handle component button linking logic toggle ("Draft Purchase Order Now") trigger hote hi calculation matrix

auto-transform ho kar pre-filled vendor procurement form execute kar dega backend tables par.

PHASE 3: DATABASE ARCHITECTURE REQUIREMENTS (AI CORE BASE TABLES)

Team Lead database setups logic define karte waqt central cloud databases aur client SQLite/IndexedDB structures mein in data structures ko expand karega:

1. **Occasion_Calendar_Matrix:** Holds localized dates boundaries mapping parameters historical reference ranges tracking coordinates ke sath.
2. **AI_Predictive_Cache_Store:** Statically caches compute calculations for stock prediction thresholds indicators taake rendering process instant non-blocking execution cycle maintain rakhe.